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ISOPLANE

Control Software for Windows

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I. Description

The Acton ISOPLANE Control Software allows the control of all Acton ISOPLANE series monochromators and some related peripheral hardware with a PC-compatible computer running Microsoft Windows. Supported versions include Windows XP, Windows Vista and Windows 7. ISOPLANE Control works on both 32 bit and 64 bit operating systems.

Functions include advancing the monochromator quickly to a wavelength, scanning at fixed speeds, changing gratings, changing turrets or diverters if installed, adjusting slit width with the motorized slit option, installing new gratings and optimizing the parameters of the monochromators. Other hardware that can be controlled includes external filter-wheels, some controllable light sources, and laser interlocks. Control for these devices will be enabled if they are detected.

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II. Installation

Any system capable of running the required operating system can run ISOPLANE Control as long as there is and at least one RS-232 COM or USB port. The installation must be done from an account with Administrative privileges.

Insert the Acton ISOPLANE Control Software Installation Disk into the CD drive of the computer. On most computers, the disk will automatically open and present the following window. If autorun is not automatically executed on your computer, then double click on the autorun file located on the ISOPLANE Control install disk from your CD ROM drive in Windows Explorer to bring up the following window.

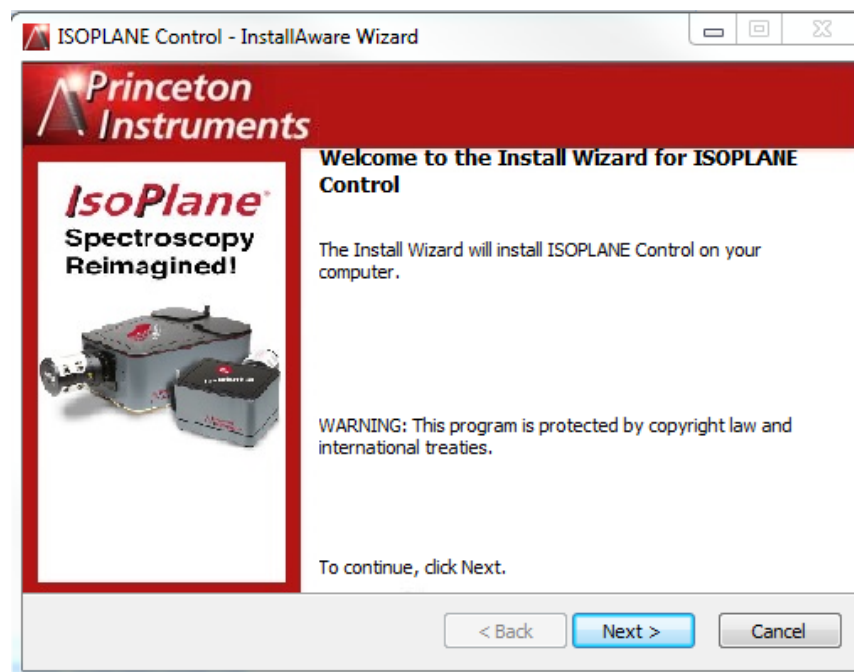


Figure 1

Click on the **Next** button and follow the instructions on the Install Wizard windows to install the ISOPLANE Control Software onto your computer.

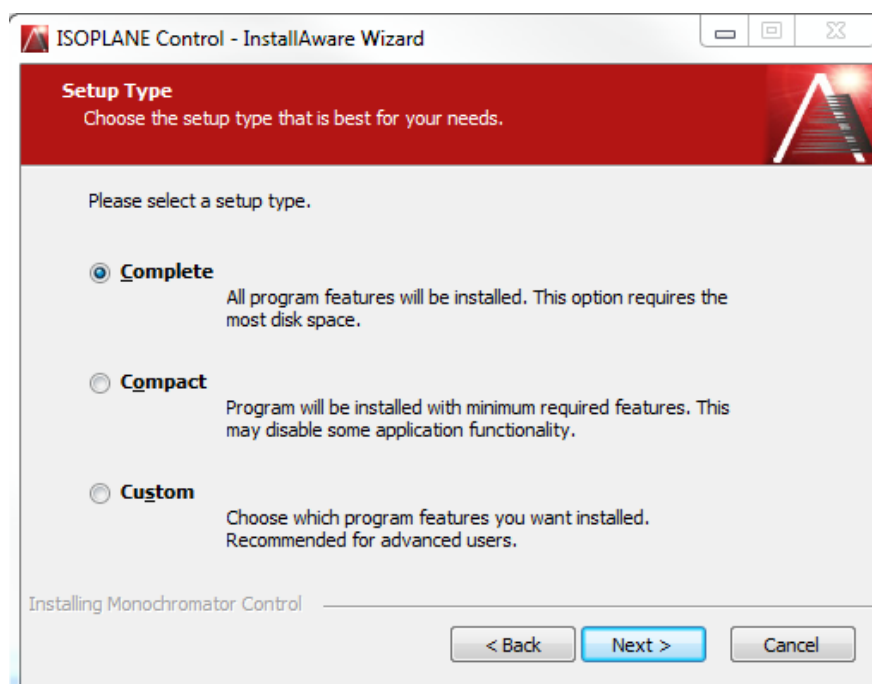


Figure 2

The necessary files for operating all of the ISOPLANE series Monochromators will be installed on the computer hard drive. The default directory for these files is C:\Program Files\Princeton Instruments\ISOPLANEControl. The ISOPLANE

Control software will be installed in this directory if **Complete** install is selected. This directory can be changed to any desired drive and directory name by clicking on **Custom** install and following the directions presented. Click on the **Next** button to continue the install process. A MonoControl icon will be placed on the desktop when install is complete.

Also included on the ISOPLANE Control Software install CD are a manual folder which contains the user manuals for each of the Acton ISOPLANE series monochromators and a SpectraPro SDK folder which contains information on the SpectraPro COM interface. The files and folders in this Software Developer's Kit describe the SpectraPro OLE COM interface functions which allow users to access the monochromator functions from their own software. Included is the complete documentation of each function with examples with a Readme file explaining the usage of this SDK.

III. User Functions

Before starting the ISOPLANE Control Software connect the RS-232 connector on the Acton ISOPLANE Series Monochromator to the Com1, Com2, or USB port of the PC-compatible computer using a serial or USB cable supplied with the monochromator or one constructed as described in the Acton ISOPLANE series Monochromator operating manual. Turn on the Acton ISOPLANE series Monochromator and allow it to initialize. The ISOPLANE Control software allows the monochromator to be controlled from either the Com1 serial port, the Com2 serial port, or the USB port. Port selection is in software and is automatically determined by which port is connected to the monochromator. It is important that the monochromator be connected to the port and be initialized before the software is started. Also, if you are using one of the serial ports, it is important that no other application have control of the serial port while the ISOPLANE Control software is operating. There will be a short delay while the parameters for a serial port are set up.

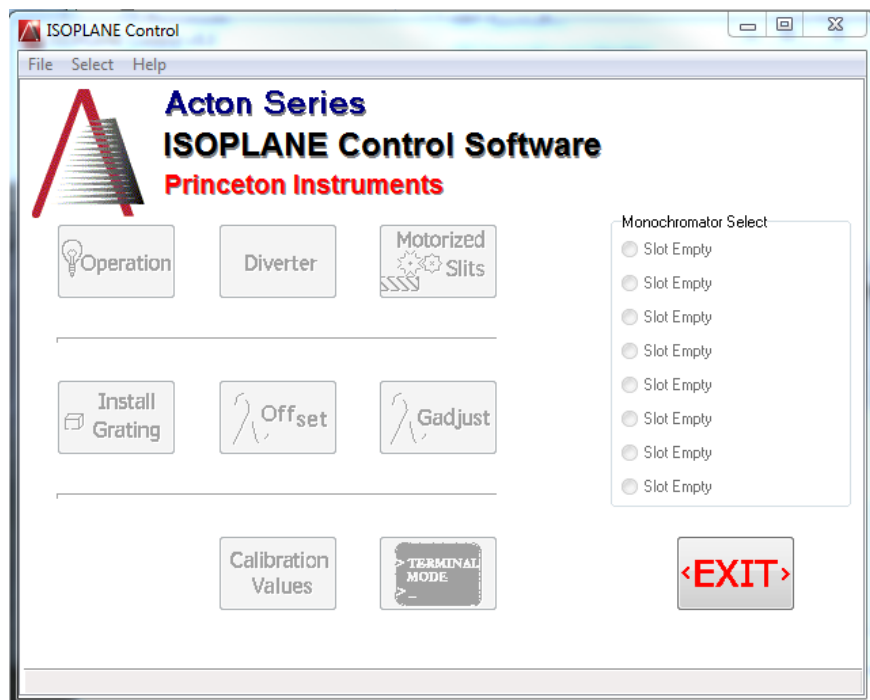


Figure 3

Figure 3 shows the main operating window for the **ISOPLANE Control Software**. All of the functions of the software are selected from this menu and control is returned to this menu when **Return** is selected in any function.

Operation:

Selecting the **Operation** button for a Monochromator allows control of the wavelength position, scan speeds and grating selection of the monochromator. When this button is clicked, the following menu is displayed.

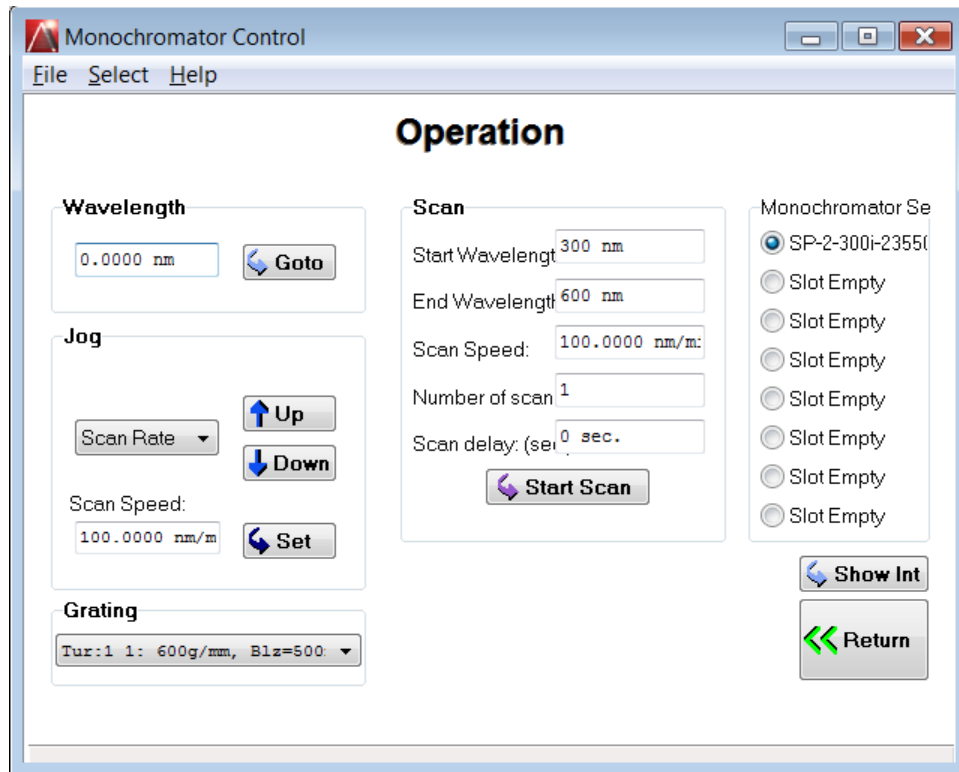


Figure 4

There are five major functions controlled from the Operation window

- GOTO a wavelength
- Jog
- Grating selection
- Turret selection
- Repeat scans

The **Wavelength** box in the **Operation** window indicates the current monochromator wavelength. Clicking the number in the **Wavelength** box will highlight the wavelength value and allow entering a new value. Values entered are in nanometers and can be to the nearest nm, nearest 0.1 nm or 0.01 nm. Clicking the **GoTo** button starts the monochromator moving at maximum speed to the selected wavelength. The display shows the new wavelength when the destination is reached.

The JOG function allows jogging the monochromator in the increasing or decreasing wavelength direction by clicking and holding down the **Up** or **Down**

buttons for as long as it is desired to move the monochromator. The jog speed is set by the three selections between the **Up** and **Down** buttons. **Maximum** is the same speed as GOTO, **Scan Rate** is the speed set in the **Scan Speed** selection under **Scan** and single step is a single motor step for a click of the **Up** or **Down** button. Note that single step will not necessarily change the wavelength display each time a click is performed.

The **Grating** section allows changing the selected grating to any of the installed gratings for the monochromator. To select an alternate grating, click the arrow beside the grating box and a dropdown selection list will appear. Double-click on the desired grating selection. Note that two gratings are shown for the SP-2150 and three gratings are shown for all other Acton ISOPLANE series monochromators.

A **Turret** selection box is provided when controlling an SP-2150, SP-2300i, SP-2500i, SP-2750i or other Acton SP series monochromators with the turret option. If a turret option is not included in the monochromator being controlled this box is not displayed. To select an alternate turret, click the arrow beside the turret box and a dropdown selection list will appear. Double-click on the desired turret selection. The grating box will change to reflect the updated turret selection.

The **Scan** box allows the monochromator to make repetitive scans. The parameters for start wavelength, end wavelength, scan speed number of scans and delay in seconds between scan are set up in the boxes provided and the scans are started with the **Start Scan** button. Again, the wavelength display box will track the wavelength of the monochromator as it is scanning.

To exit the **Operation** window and return to the **Main** menu, click in the **Return** box near the bottom of the window.

If your monochromator has either a Shutter configured or Filter attached additional relevant control options will appear on the Operations Screen:

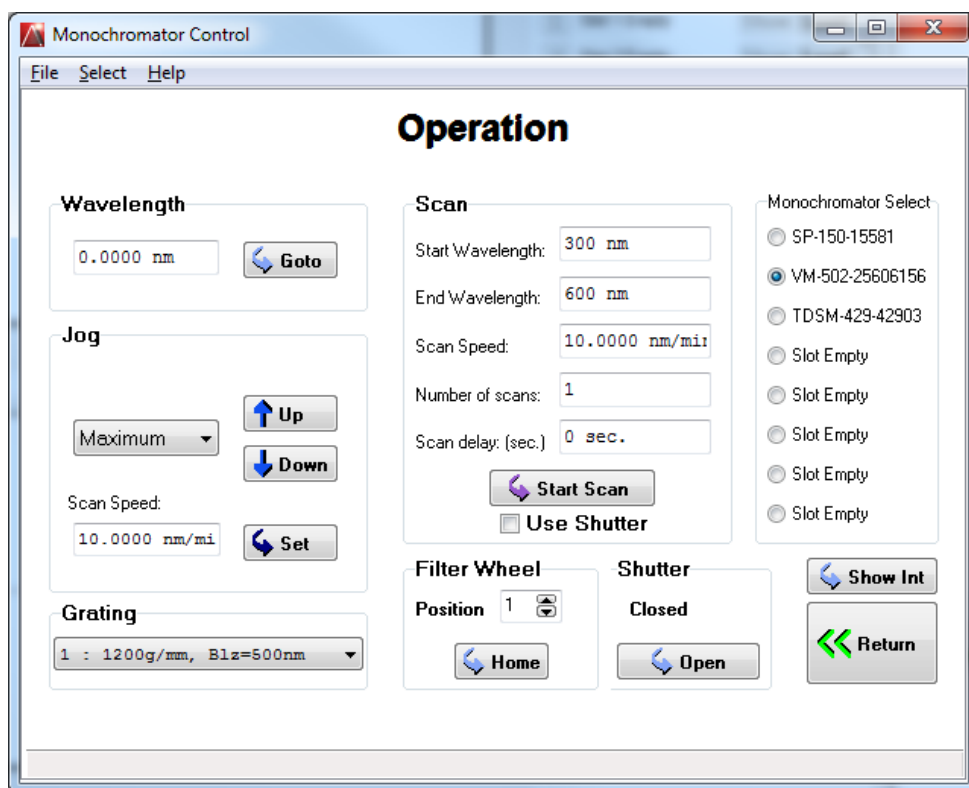


Figure 5

In addition to routine operation the Operation Screen can be used to test the general hardware functionality using the interrupter display by using the Show Int. button. Once a device has been selected this can be enabled to watch the status of the interrupters during routine operation.

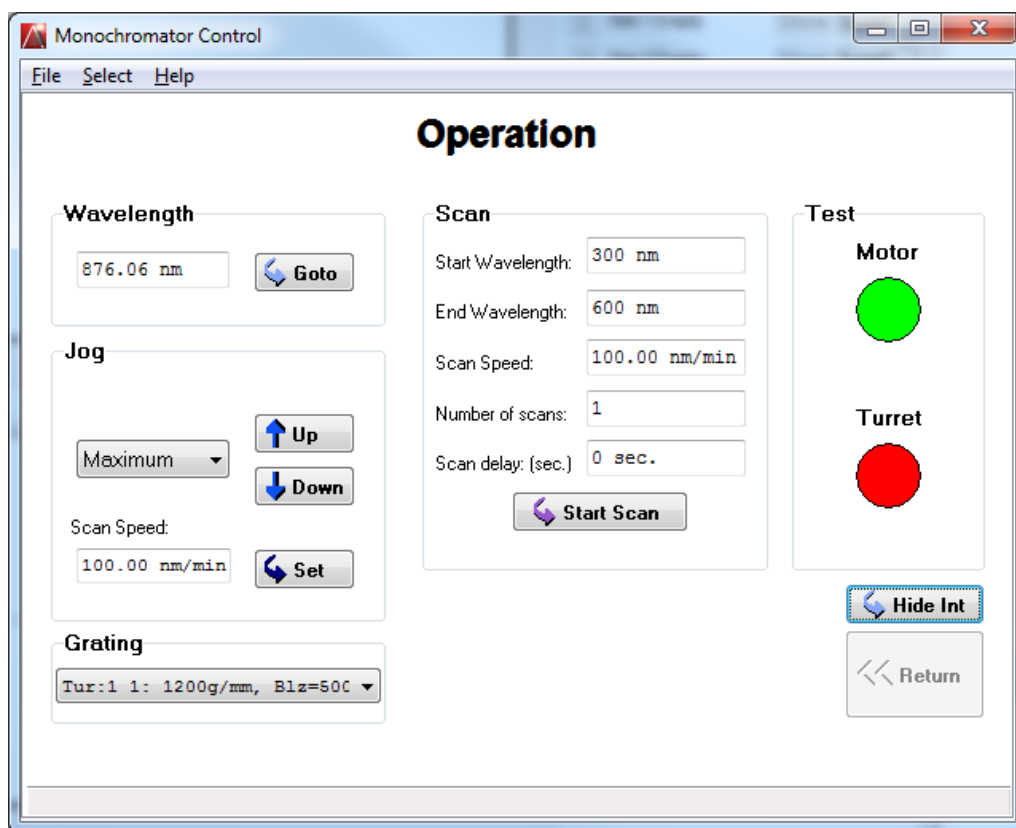


Figure 6

Filter Operation:

For a Filter the Operation button brings up a filter-specific operation screen:

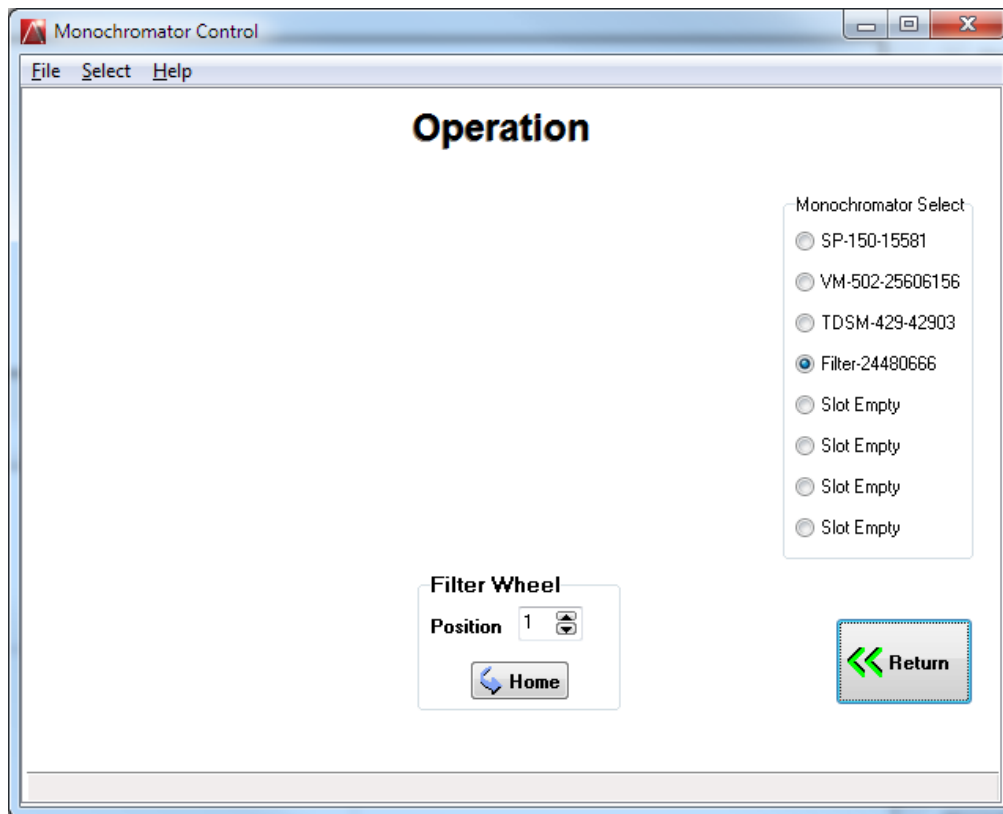


Figure 7

An external filter has been pre-programmed with a number of positions and these can be set for the filterwheel. No additional options are available for an external filter.

Light Source Operation:

For the Controllable Light Source, in this case the Tungsten Deuterium TDSM-429 the operation screen also brings up a limited operation screen that is appropriate:

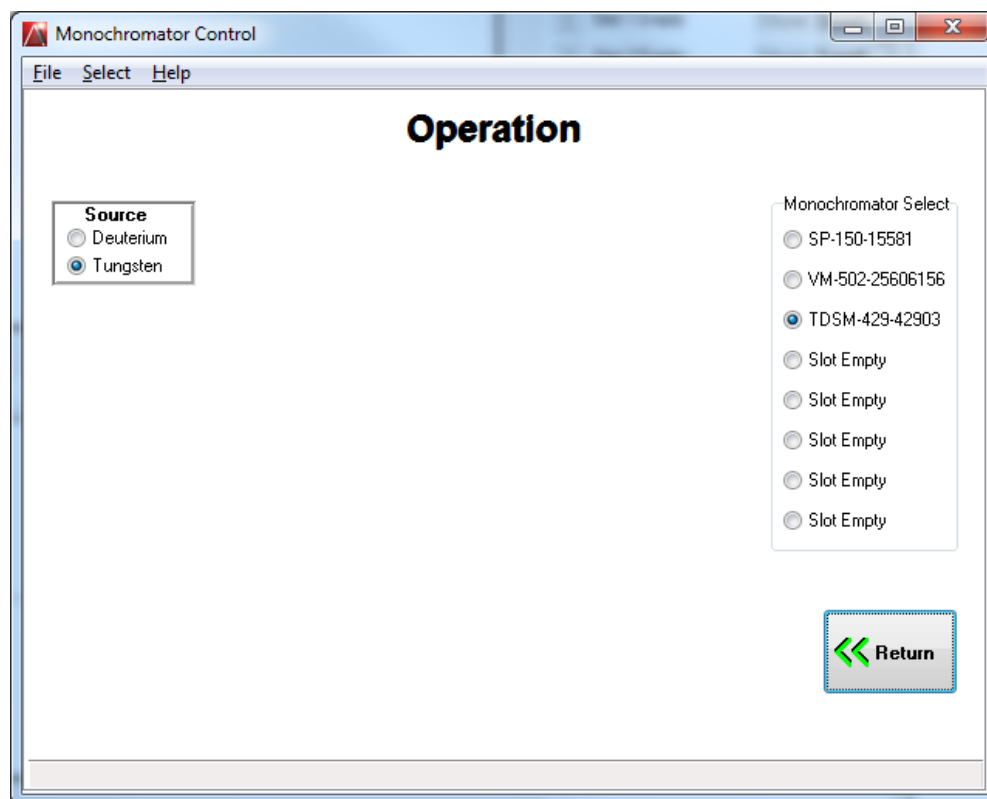


Figure 8

Diverter Mirror Control:

The **Diverter** selection button is to the right of the **Operation** button on the main window. Clicking on this button opens the window shown in Figure 5.

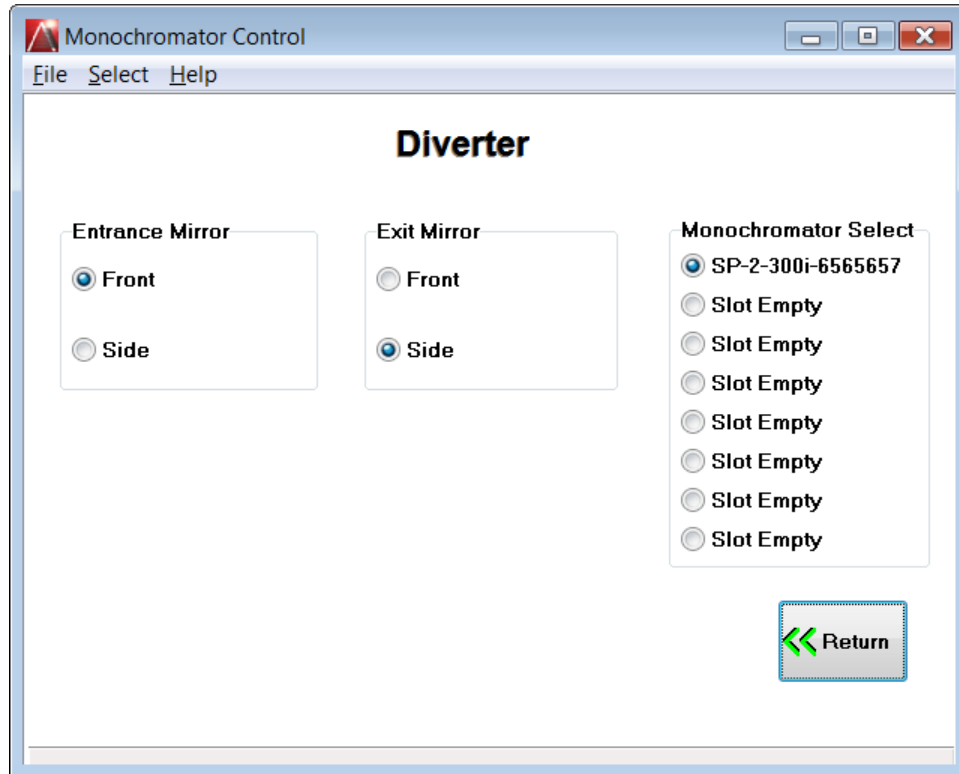


Figure 9

Clicking in the radio button beside **Front** or **Side** for any of the displayed diverter controls will move the diverter mirror to the selected position. If only one diverter is present, only that active diverter control will be displayed. If the monochromator is configured as a '**double**', with one monochromator controlling the other directly, than the controls for controlling the ports on the second monochromator will also be visible on this page.

Click the **Return** button to return to the **Main** menu.

Slit Width Control:

Clicking the **Motorized Slits** button opens the window shown below for controlling the position of each of the installed motorized slits.

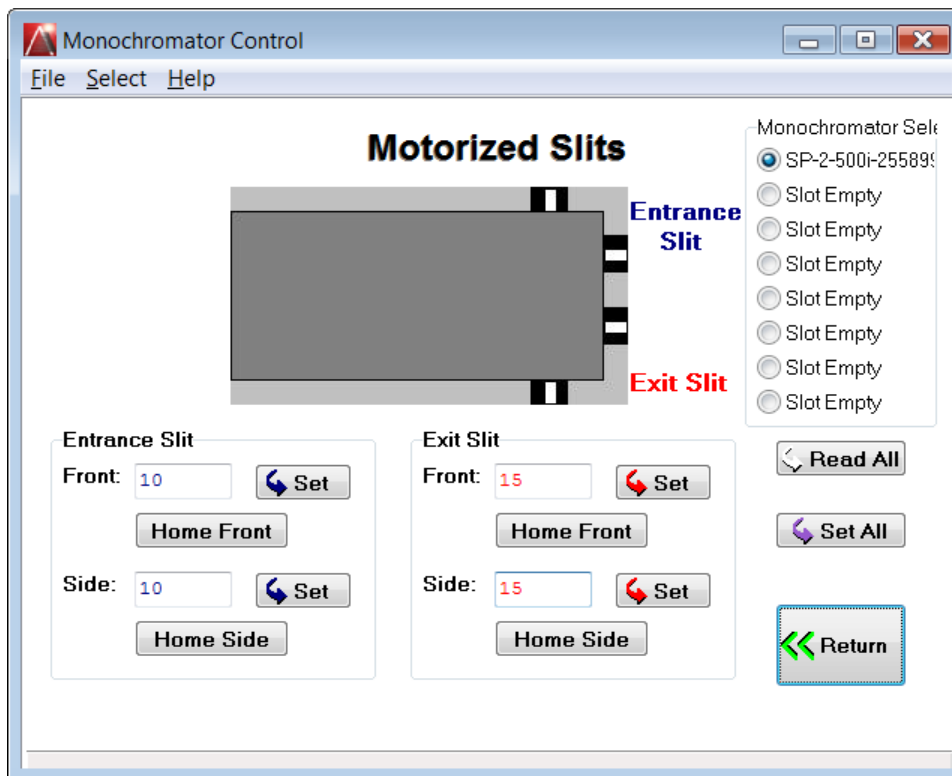


Figure 10

Controls for only the motorized slits installed in the monochromator will be displayed. For each active motorized slit, you can control of the slit width by double-clicking in the box which displays the width for the desired slit and entering a new value in increments of 1 micron. The range of the slit width is 10 microns to 3000 microns for standard motorized slits. If using an optional 12 mm slit, the range is 10 microns to 12000 microns in 5 micron increments. To make the slit go to the new value, click the **Set** button. The width of the slit is not affected by initialization of the monochromator or by turning the power off or on. Clicking the **Home Front** or **Home Side** button will cause the selected slit to initialize to its home position of 10 microns.

Click the **Return** button to return to the **Main** menu.

Install Grating:

Below the **Operation** button in the main window is the **Install Grating** button. Clicking this button opens the window shown in Figure 5 and allows the installation of additional gratings to the Controller.

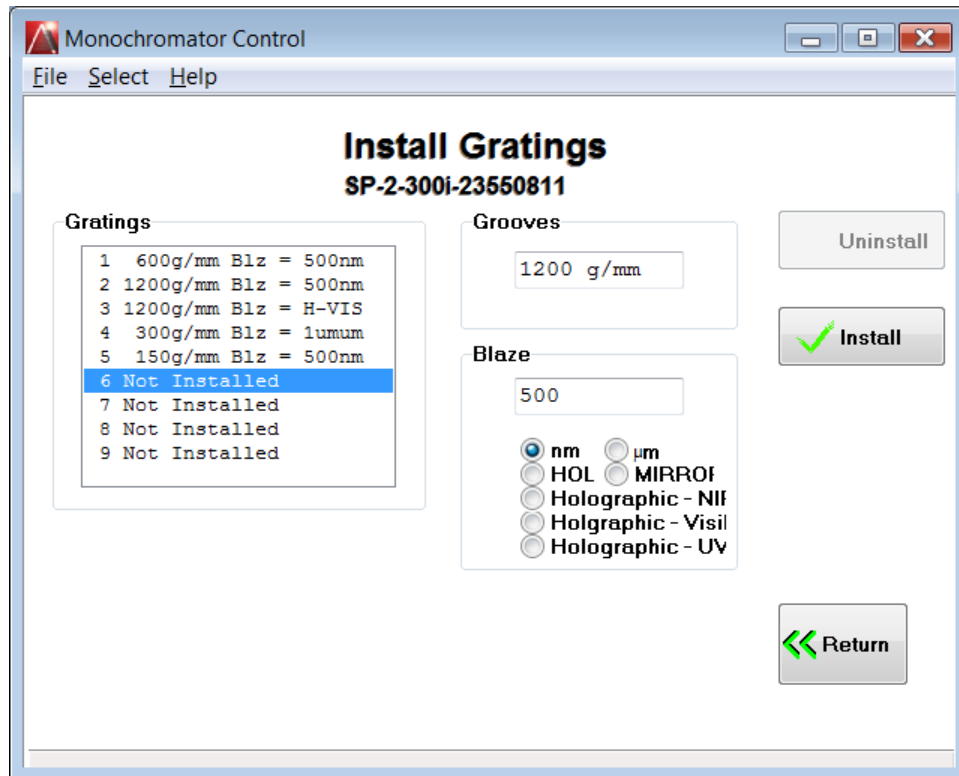


Figure 11

Note: Before installing a new grating it is advised that the monochromator parameters as shipped be saved in a file. This can be done in the **Calibration Values** window. Refer to page 21 of this manual for instructions.

Select the grating position in which the new grating is to be installed by clicking the desired grating position in the **Gratings** list. When a monochromator with replaceable turrets is used, position 4 is grating #2 in turret #2 for an SP-2150 and is grating #1 in turret #2 for a SP-2275, SP-2500 or SP-2750. Enter the number of grooves per millimeter in the **Grooves** box to the right. The number of grooves can range from 20 to 3600; no decimal points are allowed in the groove box. Enter the blaze wavelength in the **Blaze** box. The blaze can contain three numbers or letters. For example 300, 1.2 and HOL are all acceptable. Select nanometers or microns for the units of numerical blazes. Click on the **Install** button to put the selected grating into the memory of the controller. The controller will automatically reset when install is clicked. If a mistake is made during the installation process or if it is desired to remove a grating from a selected position,

click on the **Uninstall** button after making the grating selection. Use the **Return** button to return to the **Main** menu.

When a new grating is installed, it is usually necessary to calibrate the monochromator in this grating position. There are two values stored in non-volatile memory which affect calibration for each grating installed in the monochromator. These values are called “Offset” and “Gadjust”. These values can be set using the **Wavelength Offset** and **Wavelength Gadjust** functions. Offset should be set first.

Wavelength Offset:

Clicking on the **λOffset** button opens the window shown in Figure 11.

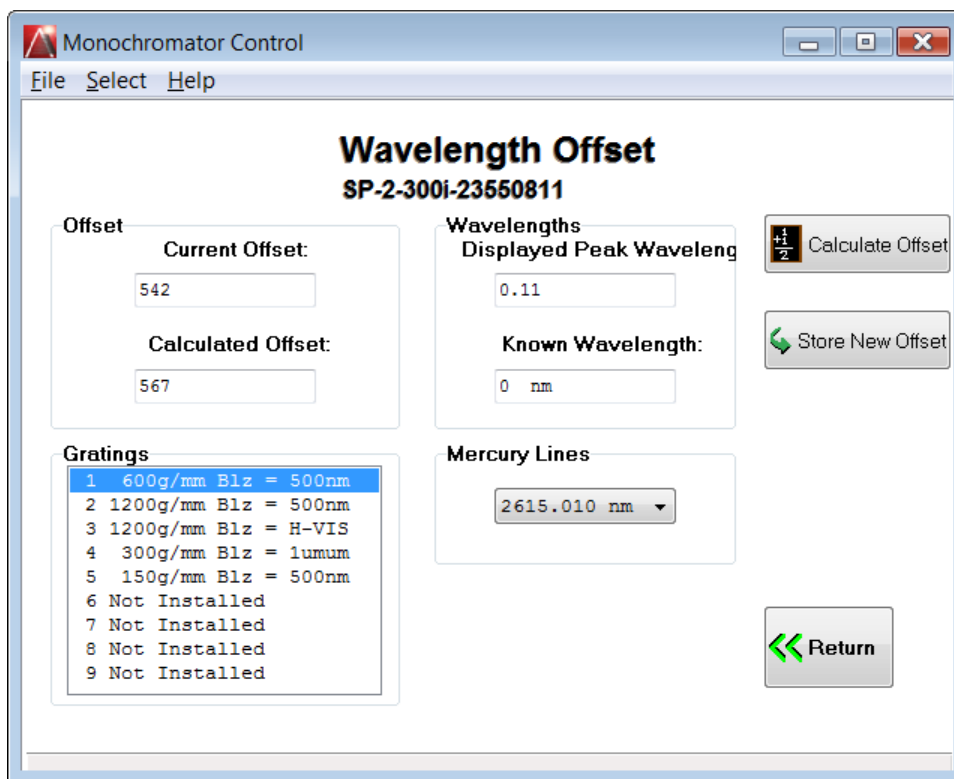


Figure 11

The **Wavelength Offset** window allows you to change the offset value for any of the installed gratings. This is typically done by finding the wavelength of peak intensity near zero wavelength using calibration source and a readout system such as the Acton NCL with SpectraSense software. If the NCL and SpectraSense are not available, any detector and readout system may be used by scanning or jogging in the operation window of this software and noting the value of the wavelength display for the wavelength that is found to have maximum intensity near zero. Return to the **Wavelength Offset** window and select the grating that is being

adjusted. The wavelength value found in the procedure above is then entered into the **Displayed Peak Wavelength** box. “0.0” is then entered into the **Known Wavelength** box. Click on the **Calculate Offset** button and the new calculated value for offset on the selected grating will be shown in the **Calculated Offset** box. If this was only a test or if the value is not one you want to save you can try other values before saving. If you wish to save the value in non-volatile memory, click the **Store New Offset** button. The monochromator will then reset and the new value of offset will be used for future monochromator activity. It is also possible to enter a value directly in the **Calculated Offset** box and click **Store New Offset**. This allows resetting a parameter to its original value if desired.

Although it is advisable to use ‘zero wavelength’ to set the offset, values other than zero wavelength may be used. It is best, however, to use low values of wavelength for adjusting the offset of a grating and high values of wavelength for setting the Gadjust.

Use the **Return** button to return to the **Main** menu.

Wavelength Gadjust:

A similar window exists for setting the “Gadjust” parameter for a grating. Clicking on the λ **Gadjust** button opens the window shown in Figure 12.

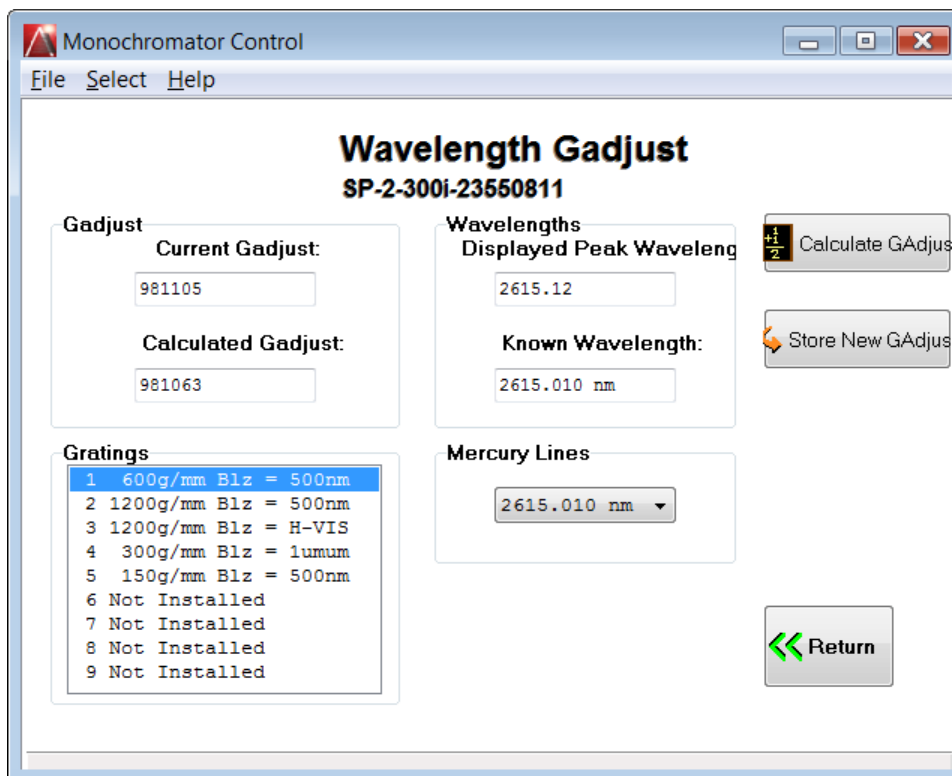


Figure 12

Again, the usual way of setting the Gadjust value is to find a known optical calibration line using a readout system and enter the wavelength value of peak intensity into the **Displayed Peak Wavelength** box. If a mercury calibration lamp is used, one of the values from the **Mercury Lines** table can be selected and this value will appear in the **Known Wavelength** box. A value can also be directly entered into the **Known Wavelength** box. Typically, a wavelength value near the high limit of the grating is used for setting the Gadjust as this gives the greatest resolution to the setting. When both wavelengths are entered, clicking the **Calculate Gadjust** button will cause a new calculated value to be displayed in the **Calculated Gadjust** box. To save this value for future use, click **Store New Gadjust**.

As before, a Gadjust value can be directly entered into the **Calculated Gadjust** box and saved with the **Store New Gadjust** button.

Use the **Return** button to return to the **Main** menu.

Startup Defaults:

Clicking on the **Startup Defaults** button opens the **Set Defaults** window shown in Figure 13.

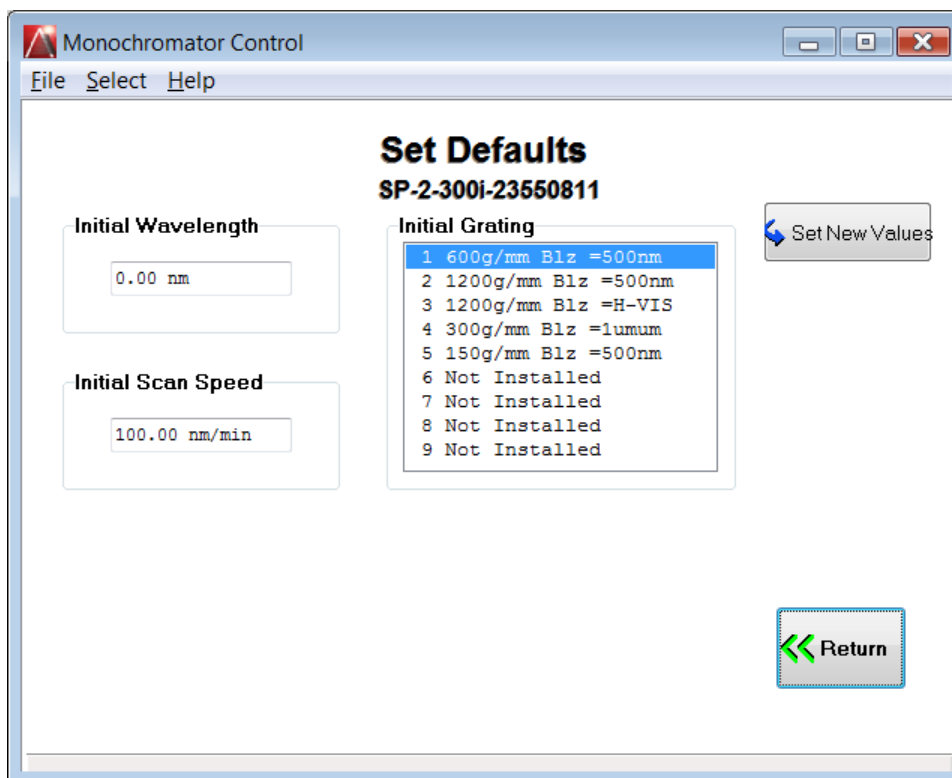


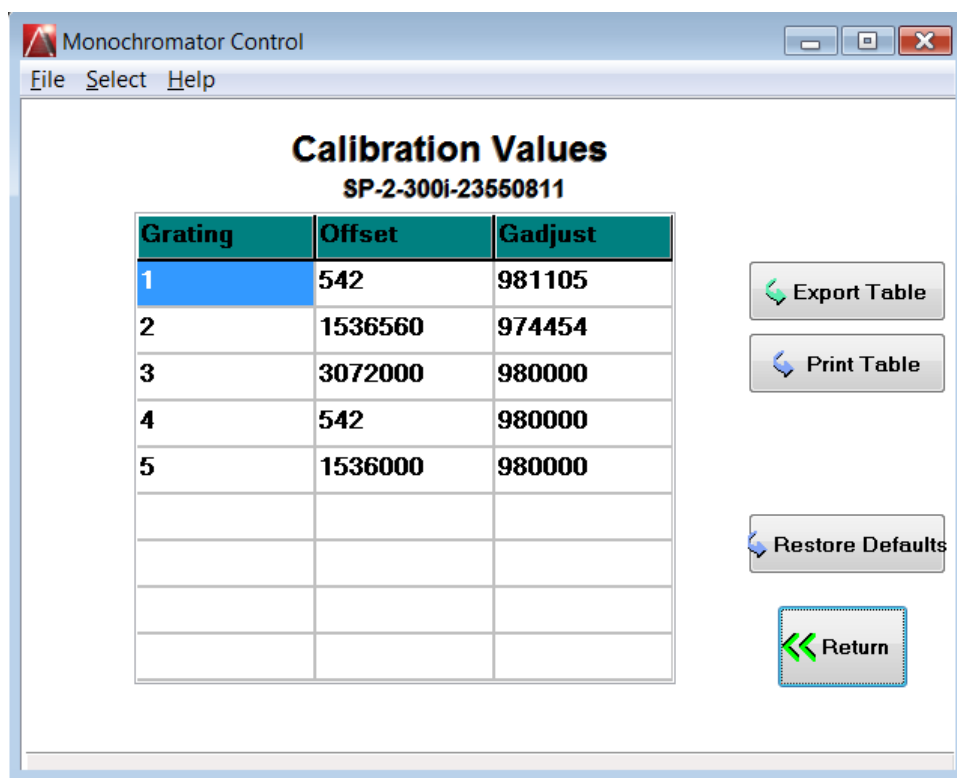
Figure 13

This window allows you to set alternate grating, wavelength and scan rate values to be used at initialization of the monochromator. The original values for these properties are Grating 1, 0.0 wavelength and 100 nm/min scan rate. If an alternate grating or wavelength value is selected to be used at initialization, the monochromator will first go to 0.0 wavelength on grating 1 and then automatically move to the selected grating and wavelength. New values may be entered into the boxes shown above and saved using the **Set New Values** button.

Use the **Return** button to return to the **Main** menu.

Calibration Values:

Clicking on the **Calibration Values** button opens window shown in Figure 14. The table displays the current state of the offset and gadjust values stored in the monochromator's non-volatile memory.



The screenshot shows a software window titled "Monochromator Control" with a menu bar containing "File", "Select", and "Help". The main area is titled "Calibration Values" with the identifier "SP-2-300i-23550811" below it. A table with three columns: "Grating", "Offset", and "Gadjust" is displayed. The first row is highlighted in blue. To the right of the table are four buttons: "Export Table", "Print Table", "Restore Defaults", and "Return".

Grating	Offset	Gadjust
1	542	981105
2	1536560	974454
3	3072000	980000
4	542	980000
5	1536000	980000

Figure 14

This window displays the calibration values set up in **Wavelength Offset** and **Wavelength Gadjust**; it does not allow setting them. There are two calibration values for each grating position. The **Offset** value is the location of zero wavelength in motor steps from the monochromator initialization point for the specified grating. The **Gadjust** value is a factor used to make fine adjustments to the higher wavelengths for a selected grating. The default values for these parameters are listed for any grating position for which a grating is not installed.

You can save a copy of the table contents in two ways. The table can be printed using the **Print Table** function or it can be saved to a file using the **Export Table** function. The information from the printed or exported table can then be used to restore any values on the **Wavelength Offset** and **Wavelength Gadjust** windows to their previous state.

To restore the values to their original state use the **Restore Defaults** button.

Use the **Return** button to return to the **Main** menu.

Terminal Mode:

Clicking on the **Terminal Mode** button opens the window shown in Figure 15. The **Terminal Mode** function allows direct control of the Acton Series monochromator using the commands described in the user manual for the monochromator.

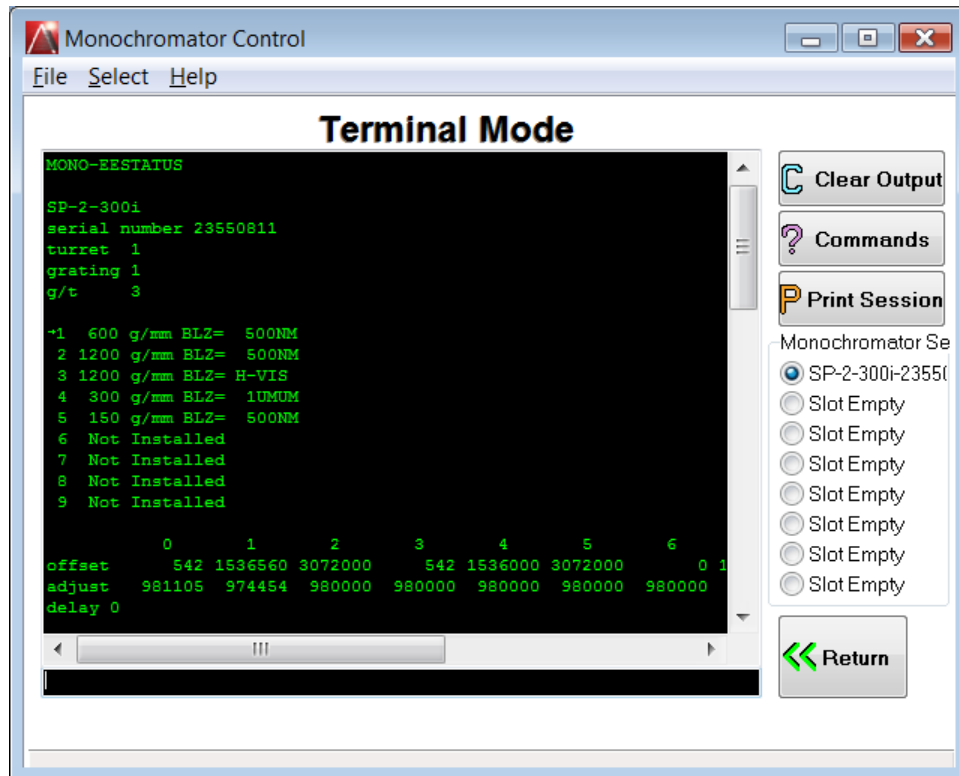


Figure 15

Note that this window only displays the values and does not allow changing them; to change the values, use the **Wavelength Offset** or **Wavelength Gadjust** windows. Before changing the values on one of these windows, it is a good idea to save a copy of the **Offset** table, located on the **Wavelength Offset** window: see page 21 for details.

Selecting **Terminal** allows direct control of the monochromator by typing in commands as described in the user manual for each monochromator. To enter a command, click in the single line near the bottom of the window and type the desired command. Note that all commands are single words – no spaces – and multiple commands may be entered on one line with spaces between the commands. Parameters are entered first followed by the command (e.g. 500.0 GOTO). Commands are not case-sensitive. Execution of the command or string of commands begins when the **ENTER** key on the keyboard is pressed. The command or string of commands is complete when “ok” is returned to the window.